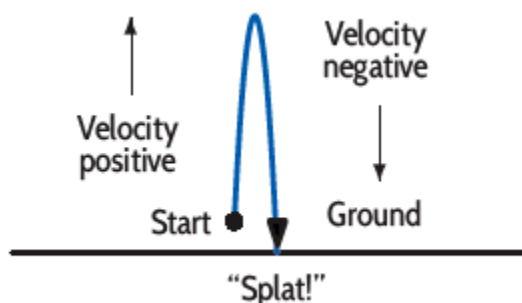


Sec 2.1 – How Do We Measure Speed?

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

What if we threw a grapefruit straight up in the air? And when it left our hand we started timing, so that would be $t = 0$ seconds. It leaves the hand at a high speed, slows down, and then for an instant when it is at its maximum height, it has no velocity, then speeds up as it comes back down.



We track the height of the grapefruit at each second after throwing:

Table 2.1 Height of the grapefruit above the ground

t (sec)	0	1	2	3	4	5	6
y (feet)	6	90	142	162	150	106	30

Can we use this table to determine the **velocity** of the grapefruit at $t = 1$ seconds?

Note that **velocity** is different than **speed**. Speed is just a magnitude, or number, while velocity also has a direction. In the example above upward is positive and downward is negative. Speed is just the magnitude of velocity, so it is always positive or zero. For example, the speedometer in your car just gives you the speed, there aren't any signs indicating direction.

Definition

If $s(t)$ is the position of an object at time t , then the **average velocity** of the object over the interval $a \leq t \leq b$ is

$$\text{Average Velocity} = \frac{\text{Change in position}}{\text{Change in time}} = \frac{s(b) - s(a)}{b - a}$$

In words, the average velocity of an object over an interval is the net change in position during the interval divided by the change in time.

Remember: Average velocity is just **slope**!

So, what is the average velocity of the grapefruit on the intervals (a) $1 \leq t \leq 3$ and (b) $4 \leq t \leq 5$?

In (a), $\frac{s(3)-s(1)}{3-1} = \frac{162-90}{3-1} = \frac{72}{2} = 36$ ft/sec (**Don't forget the units**).

In (b), $\frac{s(5)-s(4)}{5-4} = \frac{106-150}{5-4} = \frac{-44}{1} = -44$ ft/sec (**Note the negative sign means the grapefruit is coming back down**).

29. Find the average velocity over the interval $0 \leq t \leq 0.2$, and estimate the velocity at $t = 0.2$ of a car whose position, s , is given by the following table.

t (sec)	0	0.2	0.4	0.6	0.8	1.0
s (ft)	0	0.5	1.8	3.8	6.5	9.6

How would you solve this problem?

Solutions:

29.

$$\left(\begin{array}{c} \text{Average velocity} \\ 0 < t < 0.2 \end{array} \right) = \frac{s(0.2) - s(0)}{0.2 - 0} = \frac{0.5}{0.2} = 2.5 \text{ ft/sec.}$$

$$\left(\begin{array}{c} \text{Average velocity} \\ 0.2 < t < 0.4 \end{array} \right) = \frac{s(0.4) - s(0.2)}{0.4 - 0.2} = \frac{1.3}{0.2} = 6.5 \text{ ft/sec.}$$

A reasonable estimate of the velocity at $t = 0.2$ is the average: $\frac{1}{2}(6.5 + 2.5) = 4.5$ ft/sec.

To find the *instantaneous* velocity, we take the average velocity over smaller and smaller intervals; that is we find

$$\frac{s(b) - s(a)}{b - a}$$

where b is getting closer and closer to a .

Definition

We write $\lim_{x \rightarrow c} f(x) = L$ if the values of $f(x)$ approach L as x approaches c .

Suppose we want to calculate the instantaneous velocity for $s(t) = t^2$ at $t=3$. We must find:

$$\lim_{h \rightarrow 0} \frac{s(3+h) - s(3)}{h} = \lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$$

Partner up

(1) Estimate that limit numerically.

h	-0.1	-0.01	-0.001	0.001	0.01	0.1
$((3+h)^2 - 9)/h$	5.9	5.99	5.999	6.001	6.01	6.1

This looks like the instantaneous velocity is approaching 6, but this is an **estimate**. To find it **exactly**, we need to calculate the limit.

(2) Calculate that limit, using the algebraic techniques we discussed from Chapter 01.

$$\frac{(3+h)^2 - 9}{h} = \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h}.$$

cancel h , giving

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} = \lim_{h \rightarrow 0} (6 + h).$$

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} = \lim_{h \rightarrow 0} (6 + h) = 6.$$

So this algebraic result agrees with our numerical estimate.